Word2Vec vs FastText

Exploring Limitations and Improvements in Word Embeddings

Objective

Explore the limitations of Word2Vec and understand how FastText addresses these issues through hands-on implementation and analysis using the Brown corpus.

Task Overview

You will train Word2Vec and FastText models on the Brown corpus with additional test sentences designed to expose Word2Vec's limitations. You will then compare the models' performance on out-of-vocabulary (OOV) words, rare words, and morphologically related words.

Required Libraries

- **gensim** - For Word2Vec and FastText implementations
- **nltk** - For accessing the Brown corpus
- **numpy** (optional) - For numerical operations

Part 1: Data Preparation

1.1 Load Brown Corpus

Load all sentences from the NLTK Brown corpus. Ensure you have downloaded the corpus using nltk.download('brown').

1.2 Add Test Sentences

Add the following test sentences to the Brown corpus. These sentences are specifically designed to expose Word2Vec limitations:

Group 1: Morphologically Related Words

- the teacher is teaching students about grammar
- she teaches mathematics at the university
- the teacher prepared teaching materials yesterday
- many teachers attend teaching conferences annually
- effective teaching requires good communication skills

Group 2: Rare Morphological Variant (appears only once)

- the unteachable student refused to learn

Group 3: Compound and Derived Words

- computational linguistics combines computer science and language
- the computation took several hours to complete
- we computed the results using advanced algorithms
- modern computers can compute complex calculations quickly

Group 4: Rare Compound (appears only once)

- the recomputation was necessary after finding errors

Group 5: Another Morphological Family

- natural language processing is fascinating
- the nature of language is complex
- naturally occurring patterns in text are important

Group 6: Rare Morphological Variant (appears only once)

- the unnaturalness of the translation was obvious

Part 2: Train Word2Vec Model

2.1 Training Parameters

Train a Word2Vec model using the Skip-gram architecture with the following parameters:

- **vector_size:** 100
- **window:** 5
- **min_count:** 1 (to keep all words including rare ones)
- **sg:** 1 (Skip-gram)
- **epochs:** 10

2.2 Save the Model

Save your trained Word2Vec model to disk for later use and submission.

Part 3: Train FastText Model

3.1 Training Parameters

Train a FastText model using the Skip-gram architecture with the following parameters:

- **vector_size:** 100
- **window:** 5
- **min_count:** 1
- **sg:** 1 (Skip-gram)
- **epochs:** 10
- **min_n:** 3 (minimum n-gram length)
- **max_n:** 6 (maximum n-gram length)

3.2 Save the Model

Save your trained FastText model to disk for later use and submission.

Part 4: Experiments and Analysis

Experiment 1: Out-of-Vocabulary (OOV) Problem

Test words (these words do NOT appear in the training data):

- teachable (related to: teacher, teaching, teaches)
- unteacher (made-up but morphologically related)
- supercomputer (related to: computer, compute, computed)
- miscomputation (related to: computation, compute)
- unnaturally (related to: natural, naturally, nature)

Task:

1. Try to get vectors for these OOV words from both Word2Vec and FastText models
2. Record which model can handle OOV words and which cannot
3. For Word2Vec, catch the KeyError exception when accessing OOV words

Question 1: Which model can handle OOV words? Explain the mechanism that allows FastText to generate vectors for unseen words.

Experiment 2: Rare Words Quality

Rare words in corpus (appear only 1 time):

- unteachable
- recomputation
- unnaturalness

Common related words:

- teacher (appears multiple times)
- computation (appears multiple times)
- natural (appears multiple times)

Task:

4. Calculate similarity scores between rare words and their common relatives
5. Test pairs: (unteachable, teacher), (recomputation, computation), (unnaturalness, natural)
6. Compare similarity scores between Word2Vec and FastText

Question 2: Which model produces better similarity scores for rare words? Why does FastText handle rare words better than Word2Vec?

Experiment 3: Morphological Relationships

Test morphological families:

- **Family 1:** teach → teacher → teaching → teaches
- **Family 2:** compute → computer → computation → computed → computational

Task:

7. Find the top 5 most similar words for 'teaching' using both models
8. Find the top 5 most similar words for 'computation' using both models
9. Analyze whether morphologically related words appear in the top results

Question 3: Does FastText group morphologically related words better? Provide specific examples from your results showing which related words appear in the top 5.

Experiment 4: Word Analogies

Test morphological analogies:

- teacher : teaching = computer : ? (expected: computing or computation)
- natural : naturally = quick : ? (expected: quickly)

Task:

10. Use the most_similar method with positive and negative parameters
11. Compare the top result from both models

Question 4: Which model performs better on morphological analogies? Explain why.